

21) Which of the following instructions converts any BCD value available in AX register into ASCII?
Hint. ASCII equivalent of a BCD code can be obtained by appending $(011)_2$ to left of the BCD code.

- A) AND AX, 07h
- B) OR AX, 0Fh
- C) AND AX, 0Fh
- D) XOR AX, 0Fh
- E) OR AX, 30h

22) Assume that SP=0100H. What is the value of SP after execution of POP AX instruction?

- A) 00FFH
- B) 0101H
- C) 0102H
- D) 00FEH

23) Which of the following communication architectures are parallel?

- I. USB
- II. PCI
- III. PCI Express
- IV. ISA

- A) I and II
- B) I, II, III
- C) I and III
- D) II and IV

24) How many (128Mx16) memory device is needed to store 1 G bytes of data?

- A) 1
- B) 16
- C) 2
- D) 8

25) Which of the following connections are common to all memory devices?

- I. Address
- II. Read Control
- III. Write Control
- IV. Data

- A) I, II, IV
- B) I, II, III, IV
- C) Only IV
- D) I, II, III, IV

1) Which type of jump instruction is used to jump to any memory location?

- I. Short
- II. Near
- III. Far
- IV. Relative

A) I and II

B) II and III

☒ C) III and IV

☒ D) Only III

E) I and IV

2) How many (128Mx16) memory device is needed to store 1 G bytes of data?

A) 2

B) 8

☒ C) 4

☒ D) 16

E) 1

3) INC instruction is a special type of addition. However, _____ flag is not affected by INC instruction unlike ADD.

A) S

B) Z

☒ C) C

D) P

E) A

4) Regarding SRAM and DRAM, which of the following statements are true?

- I. DRAM is faster than SRAM.
- II. SRAM is more expensive than DRAM.
- III. DRAM requires periodical refresh.
- IV. SRAM consumes more power than DRAM.

☒ A) Only I

☒ B) III and IV

☒ C) I, II, IV

D) I and IV

☒ E) II and III

5) Which of the following flag bits is not tested by the conditional jump instructions?

A) S

B) C

☒ C) D

☒ D) O

E) P

6) TEST instruction performs _____ operation that affects the flags register.

A) OR

☒ B) NOR

C) XOR

D) NOT

☒ E) AND

7) Which of the following flags indicates whether the result of an arithmetic operation is positive or negative?

☒ A) S

B) C

C) O

D) Z

E) D

1) Instruction Queue is placed in _____ of Intel 8086.

- A) Control and Timing Unit
- B) Execution Unit
- C) Register File
- D) Bus Interface Unit
- E) Arithmetic Logic Unit

2) _____ is the act of synchronizing two systems that op

- A) Serial communication
- B) Handshaking
- C) Transmission
- D) Parallel communication
- E) Peripheral interface

3) Which of the followings are interrupt pins found on Intel micro

- I. INTA
- II. INTC
- III. INTQ
- IV. INTR

A) I and IV

B) I, III, IV

C) Only III

16) In 8086, 32-bit addition can be performed by adding low 16-bits of the operands and high 16-bits of the operands using _____ instruction.

- A) ADC - SBB
B) ADD - ADC
C) ADC - ADC
D) ADD - ADD
E) ADC - ADD

17) What is the IEEE 754 floating point equivalent of $-\infty$?

- [illegible]

18) It is given that the following code computes factorial of var_N. Then, what code?

```
MOV AX, 1
MOV BX, var_N
MOV CX, var_N
label:
MUL BX
missing instruction?
LOOP label
RET
var_N DW 6
```

- A) INC AX B) DEC CX C) INC BX

19) Which field of an instruction specifies the addressing mode to be used?

- A) REG B) R/M C) Direction

20) Assume that CS=1000H, DS=2000H, SS=3000H, and IP=2000H. Then next instruction executed by the microprocessor?

- A) 12000H B) 22000H C) 1000H

4) Which of the following communication architectures are serial?

- I. USB
- II. PCI
- III. PCI Express
- IV. ISA

A) I, II, IV B) I, II, III C) I and III D) Only I

5) Which flag is tested by a conditional interrupt instruction?

A) Interrupt B) Carry C) Parity D) Overflow

6) Regarding SRAM and DRAM, which of the following statements are true?

- I. DRAM is faster than SRAM.
- II. SRAM is more expensive than DRAM.
- III. DRAM requires periodical refresh.
- IV. SRAM consumes more power than DRAM.

A) II and III B) I, II, IV C) Only I D) I and IV

7) What is the IEEE 754 single precision floating point representation of decimal +10.0?

A) 41200000h B) C1A00000h C) 41A00000h D) C1200000h

8) Which of the following statements is not true when an interrupt occurs?

- I. Flags register contents are pushed on the stack.
- II. Interrupt flag is set.
- III. Contents of CS and IP are pushed onto the stack.
- IV. Related ISR is executed.

A) Only II B) I and III C) III and IV D) II and IV

9) Using little endian format, 4321h is to be stored in an 8-bit memory starting at 200h memory address 201h after storing?

A) 34h B) 43h C) 12h D) Empty

8) Which of the following memory types are volatile?

- I. SRAM
- II. DRAM
- III. Flash
- IV. EPROM

A) I, II, IV

B) III and IV

☒ C) I and II

D) Only IV

☒ E) I, II, III, IV

9) A short jump allows a program to branch forward _____ bytes or backwards _____ bytes from the next instruction's address in the program.

☒ A) 127, 128

B) 128, 127

☒ C) 256, 255

~~D) 255, 256~~

~~E) 32K, 32K~~

10) 12-bit Hamming coded data including 8-bit actual data and 4-bit parity is received as given below.

Bit	1	2	3	4	5	6	7	8	9	10	11	12
Data	0	0	1	1	1	0	0	1	0	0	0	0

Using the error checking method provided below, detect position of the error (if any) within the code.

C1 = XOR of bits (1, 3, 5, 7, 9, 11)

C2 = XOR of bits (2, 3, 6, 7, 10, 11)

C4 = XOR of bits (4, 5, 6, 7, 12)

C8 = XOR of bits (8, 9, 10, 11, 12)

A) Bit-2

B) Bit-6

C) No error

☒ D) Bit-10

E) Bit-4

Using the error checking method provided below, detect position of the error

- C1 = XOR of bits (1, 3, 5, 7, 9, 11)
- C2 = XOR of bits (2, 3, 6, 7, 10, 11)
- C4 = XOR of bits (4, 5, 6, 7, 12)
- C8 = XOR of bits (8, 9, 10, 11, 12)

- A) Bit-2 B) Bit-4 C) Bit-6 D)

11) What is the corresponding value of -76.5 in nearest even rounding mode?

- A) -77 B) -75 C) -78 D)

12) The code below computes the sum of numbers from 1 to N using the for instruction. Which instruction should be inserted into the missing line within the code?

$$\sum_{x=1}^N x = N.(N+1)/2$$

```
ORG 100h  
CALL sum  
RET
```

```
sum PROC  
MOV AL, N  
MOV BL, N  
INC BL  
MUL BL  
missing instruction?  
RET  
sum ENDP
```

N DB 8

- A) DEC AL B) SHR AL, 1 C) SUB AL, 2

13) What is the decimal equivalent of IEEE 754 single precision floating point number 0xC1900000?

- A) -20.25 B) -40.125 C) -10.50

14) What is the size of Personal Computer I/O map?